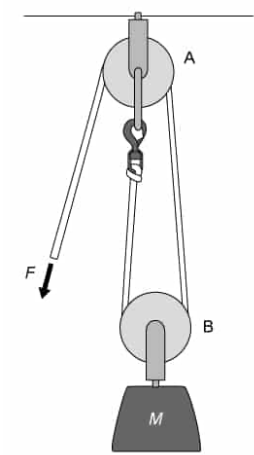


The suspension of mass M within a two-pulley system requires a force F .

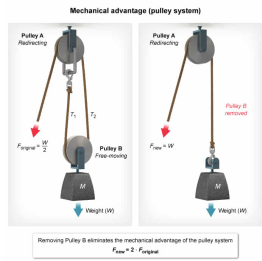


If pulley B is removed from the system, what is the ratio of the original value of the force with two pulleys to the new value of the force with one pulley?

- ☐ A. 1:1 [12%]
- ☒ B. 1:2 [68%]
- ☐ C. 1:3 [9%]
- ☐ D. 1:4 [10%]

Correct
68% answered correctly

Explanation:



Mechanical advantage describes the phenomenon by which forces are **amplified** using a mechanical device or system. One way to achieve mechanical advantage is by using a **pulley system** comprising a rope and one or more **pulleys** (ie, a wheel located on an axle) that may be configured to significantly reduce the force required to transport or support the **weight (W)** of an object.

Pulley systems can include both fixed and load-sharing pulleys. **Fixed pulleys** do not contribute to mechanical advantage but redirect forces acting along the vertical axis. For example, an upward force may be converted into a downward force as a rope bends around a fixed pulley attached to the ceiling. In contrast, **load-sharing pulleys** are typically free to move up and down in accordance with the motion of the supported object, and the mechanical advantage of a pulley system is related to the number of load-sharing pulleys that are present.

In this question, a force (F_{original}) is required to support the weight of mass M using a two-pulley system. Pulley A is fixed to the ceiling and therefore redirects the tension within the portion of the rope held by the user. Conversely, pulley B moves freely and acts to split the weight of the mass into two equal tension forces (T_1 and T_2). Tension force T_1 acts on the portion of the rope attaching pulley B to the ceiling, while tension force T_2 acts on the portion of the rope between pulley B and the user such that:

$$F_{\text{original}} = T_2 = \frac{W}{2}$$

As a result, the inclusion of pulley B within the system yields a mechanical advantage of two. Consequently, removing pulley B from the system will double the magnitude of force required to support the weight of the mass.

$$F_{\text{new}} = 2 \cdot F_{\text{original}} = W$$

Therefore, the ratio of the original force to the ratio of the new force (F_{new}) is 1:2 (**Choice B**).

Educational objective:

Mechanical advantage describes the force amplification that occurs through the use of a mechanical system or device. The mechanical advantage of pulley systems is related to the number of load-sharing pulleys but unrelated to the number of fixed pulleys that redirect forces.

Time spent:0 sec
Subject:
Physics

QID:402487
System:
Mechanics and Energy